Date : 16.7.26 EXP: 2

Study of different types of Network Cables and crimping of Cables with RJ45 Connector

Date : 23/7/26
Exp = No: 2
Aim :
To study the different types of network

a/ Understanding different types of network cables
The different types of cables used in network

1. unsheduled twisted pair (UTP) cables
2. ~~sche~~ coaxicll cable
3. fibre optic Cables

Cable type	Category	maximum data trasmission	Advandage / Disadvantage	Application
UTP	Category 3 Category 5 Category 5e	10 bps upto 100 mbps 1Gpps	Advandages → Cheaper → entry in install Disadvantages → more prone to EMI & noise	10 Base T ethernet Fast ethernet
STP	Category 6, 6a	10 Gbps	Advandages → sheilded → faster then UTP → less to noise	Gigabit ethernet

cable type	Category	maximum data transmission	Advantage/ disadvantage	Application use
Coaxial cable	12G-6 RG-59 RG-11	10-100 Mbps	Advantage → High bandwidth → versatile disadvantage → limited distance → cost → size in bulky	Speed of signal is 500 m, television network
Fibre Optic cable	Sigle mode multimode	100 Gbps	Advantage → High speed → High bandwidth → High security disadvantage → Expensive → require skilled installers	maximum distance of fibre optic cable is 100 meter

Result :-

Thus the different types of network cables were studied and verified.